

Honors Calculus III, Homework 1

Please upload solutions to **Problem 9 and Question 2** by **8pm on Friday, March 31**. You can use any results you want from lectures, provided you state them clearly, but make sure to write complete proofs. The remaining problems will not be graded, and you should not hand them in, but you are still expected to learn how to do them, unless it says “optional”.

Important: unless stated otherwise, to get full credit you need to show and explain your work. This applies both to homework and exams.

1. Spivak, *Calculus*, Chapter 24: Problems 1, 2, 3, 5, 9(*5pt), 16, 29.
2. If $n \in \mathbf{N}$ and $\alpha \in \mathbf{R}$ we define the binomial coefficient by the formula

$$\binom{\alpha}{n} = \frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!},$$

which generalizes the usual definition when $\alpha \in \mathbf{N}$. For the rest of this problem we fix some $\alpha \in \mathbf{R}$.

- (a) (*5pt) Prove that

$$f(x) = \sum_{n=0}^{\infty} \binom{\alpha}{n} x^n$$

converges absolutely if $|x| < 1$.

- (b) (*5pt) Prove that $(1+x)f'(x) = \alpha f(x)$ whenever $|x| < 1$.
(c) (*5pt) Prove that $f(x) = (1+x)^\alpha$ whenever $|x| < 1$.

Solutions to Problem 9. We will use the Weierstrass M -test. To find the numbers M_n , we compute the maximum and the minimum of

$$f_n(x) = \frac{x}{n(1+nx^2)}$$

on $\mathbf{R}$. Notice that f_n is differentiable infinitely many times on $\mathbf{R}$, since $1+nx^2 > 0$ for all $x \in \mathbf{R}$. The derivative is

$$f'_n(x) = \frac{n - n^2x^2}{n^2(1+nx^2)^2}$$

which vanishes at $x = \pm 1/\sqrt{n}$. We claim that $1/\sqrt{n} = \sup_{x \in \mathbf{R}} |f(x)|$. (Any complete argument gets full points, regardless of whether it is the same as the one below, but you do need to justify this.) To see this, notice that f'_n is positive on $(0, 1/\sqrt{n})$ and negative on $(1/\sqrt{n}, +\infty)$. If $x \in (1/\sqrt{n}, +\infty)$ then the mean value theorem implies that

$$f(x) - f(1/\sqrt{n}) = f'(t)(x - 1/\sqrt{n})$$

for some $t \in (1/\sqrt{n}, +\infty)$. Hence $f(x) > f(1/\sqrt{n})$. By the same argument, this inequality is also true for $x \in (0, 1/\sqrt{n})$, hence $1/\sqrt{n}$ is the maximum value of f on $(0, +\infty)$. Since f_n is an odd function, this implies that $1/\sqrt{n} = \sup_{x \in \mathbf{R}} |f(x)|$.

It follows that

$$\sup_{x \in \mathbf{R}} |f(x)| = \sup_{x > 0} |f(x)| = f(1/\sqrt{n}) = \frac{1}{2n\sqrt{n}}.$$

Defining $M_n = 1/2n\sqrt{n}$, we see that we are done since the series

$$\sum_{n=1}^{\infty} \frac{1}{2n\sqrt{n}}$$

converges, by a result from Math 16200.

Solutions to Question 2. (a) This is immediately seen to be true if $x = 0$, and if $0 < |x| < 1$ then it follows from the ratio test. More precisely, we compute that

$$\left| \binom{\alpha}{n+1} x^{n+1} \binom{\alpha}{n}^{-1} x^{-n} \right| = \left| \frac{\alpha - n}{n+1} x \right|$$

which tends to $|x| < 1$ as $n \rightarrow \infty$. Hence the series

$$\sum_{n=0}^{\infty} \left| \binom{\alpha}{n} x^n \right|$$

converges by the ratio test.

(b) Choose any $x_0 \in (|x|, 1)$. Since $f(x_0)$ converges, by a result in class we deduce that f is differentiable at x and

$$f'(x) = \sum_{n=0}^{\infty} (n+1) \binom{\alpha}{n+1} x^n.$$

It follows that

$$(1+x)f'(x) = \sum_{n=0}^{\infty} \left((n+1) \binom{\alpha}{n+1} + n \binom{\alpha}{n} \right) x^n = \sum_{n=0}^{\infty} \left(\alpha \binom{\alpha-1}{n} + \alpha \binom{\alpha-1}{n-1} \right) x^n$$

and so we are done since

$$\binom{\alpha-1}{n} + \binom{\alpha-1}{n-1} = \binom{\alpha}{n}.$$

(Make sure you know how to prove this even if α is not an integer.)

(c) We need to prove that $h(x) = f(x)(1+x)^{-\alpha}$ is constantly equal to 1. (Notice that this function is well-defined since $1+x > 0$ whenever $|x| < 1$.) We compute the derivative:

$$h'(x) = f'(x)(1+x)^{-\alpha} - \alpha f(x)(1+x)^{-\alpha-1} = 0$$

by part (b). Since the domain of $h(x)$ is an interval, it follows that $h'(x)$ is constant. Evaluating at $x = 0$ we find that $h(0) = 1$, and so $h(x) = 1$ for all $x \in (-1, 1)$.